

NUPOOR GANDHI

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EDUCATION

Carnegie Mellon University

August 2020 - Present

PhD Student, Machine Learning and Public Policy

Pittsburgh, PA

Advisor: Emma Strubell

Thesis: *Center-based Decomposition for Efficient Annotation of Structured Representations*

University of Illinois at Urbana-Champaign

August 2016 - May 2020

Bachelor of Science, Computer Science

Minor in Mathematics

Urbana, IL

Advisor: ChengXiang Zhai

Senior Thesis: *Forecasting Public Health Outcomes with Social Media Data*

PUBLICATIONS

N. Gandhi, E. Strubell, Task Decomposition for Efficient Annotation, Journal of Computational Linguistics 2026 (Under Review)

N. Gandhi, M. Bada, E. Strubell, Decomposing Unitization and Typing for Efficient and Consistent Span-Bound Concept Annotation, Findings of ACL 2026

K. Ramesh, D. Smolyak, Z. Zhao, **N. Gandhi**, R. Agarwal, M. Bjarnadóttir, A. Field, **SynthTextEval: Synthetic Text Data Generation and Evaluation for High-Stakes Domains**, Proceedings of EMNLP 2025 - Demo

S. Gururaja, **N. Gandhi**, J. Milbauer, E. Strubell, **Beyond Text: Characterizing Domain Expert Needs in Document Research**, Findings of ACL 2025

K. Ramesh, **N. Gandhi**, P. Madaan, L. Bauer, C. Peris, A. Field, **Evaluating Differentially Private Synthetic Data Generation in High-Stakes Domains**, Findings of EMNLP 2024

N. Gandhi, T. Corringham, E. Strubell, **Challenges in End-to-End Policy Extraction from Climate Action Plans**, Proceedings of ClimateNLP Workshop at ACL 2024

N. Gandhi, A. Field, E. Strubell, **Mention Annotations Alone Enable Efficient Domain Adaptation for Coreference Resolution**, Proceedings of ACL 2023

A. Field, A. Coston, **N. Gandhi**, A. Chouldechova, E. Putnam-Hornstein, D. Steier and Y. Tsvetkov, **Examining risks of racial biases in NLP tools for child protective services**, Proceedings of ACM FAccT 2023

N. Gandhi, A. Field, Y. Tsvetkov, **Improving Span Representation for Domain-adapted Coreference Resolution**, Proceedings of the EMNLP Fourth Workshop on Computational Models of Reference, Anaphora and Coreference, 2021

N. Gandhi, A. Morales, S. Chan, D. Albarracín and C. Zhai, **Predicting Opioid Abuse with text-based Twitter Features**, Proceedings of the Thirty-Fourth AAAI Conference on Artificial Intelligence, 2020 (*Student Abstract*)

A. Morales, **N. Gandhi**, S. Chan, “**Multi-Attribute Topic Feature Construction for Social Media-based Prediction**,” 2018 IEEE International Conference on Big Data (Big Data), Seattle, WA, USA, 2018, pp. 1073-1078.

TEACHING	TA, Computational Ethics in NLP (11-830) Language Technologies Institute, CMU	January 2022-May 2022
	TA, Probabilistic Graphical Models (10-708) Machine Learning Department, CMU	January 2026-May 2026
SERVICE	<i>Reviewer</i> , ACL Rolling Review (2026) <i>Reviewer</i> , ACL - Climate NLP Workshop (2024-2026) <i>Reviewer</i> , Journal of Comparative Policy Analysis (2024) <i>Reviewer</i> , EMNLP (2023) <i>Application Reviewer</i> , Data Science for Social Good Fellowship (2022, 2026) <i>Mentor</i> , CMU AI Mentoring Program (2021 - present) <i>Committee Member</i> , Machine Learning Department Masters Admissions (2022)	
OTHER ACTIVITIES	<i>GSA Representative</i> , Graduate Student Assembly (2023 - present) <i>Student Leadership Committee Representative</i> , CMU Senate (2025-2026) <i>Volunteer</i> , Scopeathon Data Science for Social Good CMU (2022, 2025) <i>Volunteer Data Scientist</i> , Illinois Legal Aid Online (2020) <i>President</i> , Society of Women Engineers (2019 - 2020) <i>External Vice President</i> , Society of Women Engineers (2017 - 2018)	
AWARDS	<i>AAAI 2020 Travel Grant</i> to attend conference <i>Knights of St. Patrick</i> , highest award for leadership by College of Engineering 2020 <i>Illinois Scholars Research Grant</i> , awarded to Illinois students for excellence in research 2017 <i>Gold Collegiate Section Award</i> , awarded at Society of Women Engineers Conference 2018 <i>Rapid Fire Research Finalist</i> , awarded at Society of Women Engineers Conference 2018 <i>Outstanding Poster Presentation - Science & Mathematics</i> , awarded at Undergraduate Research Week 2019	
TALKS	Challenges in End-to-End Policy Extraction from Climate Action Plans <i>Poster Presentation at ClimateNLP Workshop ACL 2024</i> <i>Oral Presentation at Climate NLP Reading Group</i>	
	Centering as a Critical Challenge for Efficient Annotation of Structured Representations <i>Poster Presentation at Midwest Speech and Language Days 2025</i>	
	LLMs for Few-shot Information Extraction with Climate Action Plans <i>Poster Presentation at Text-as-Data Conference 2023</i>	
	Risk Prediction and Information Extraction for Child Welfare in Allegheny County <i>Guest Lecture for Computational Ethics in NLP 2022</i>	
	Mention Annotations Alone Enable Efficient Domain Adaptation for Coreference Resolution <i>Oral Presentation at ACL 2023</i>	
	Improving Span Representation for Domain-adapted Coreference Resolution <i>Oral Presentation at EMNLP 2021 CRAC Workshop</i>	
	Reducing response times to citizen legal questions in Uganda <i>Lightning Talk at KDD 2020 — Social Impact Session</i> <i>Oral Presentation at Data Science for Social Good: DataFest 2019</i>	

Predicting Opioid Abuse with text-based Twitter Features
Poster Presentation at AAAI Conference on Artificial Intelligence 2020

Natural Selection and the Bystander Effect
Poster Presentation at Undergraduate Research Symposium 2019

Multi-dimensional Features for Prediction with Tweets
Poster Presentation at ACM Tapia Conference 2018

Predicting Sexually Transmitted Infections with Twitter Data
Rapid-Fire talk at Society of Women Engineers Conference 2018
Poster Presentation at Illinois Scholars in Undergraduate Research Expo 2018

WORK

Data Science for Social Good Fellow, Barefoot Law
Imperial College London June 2019 - August 2019
Advisor: Professor Rayid Ghani, Professor Leid Zejnilovic

- Built an Information Retrieval-based response recommender system to help Barefoot-Law lawyers to reach citizens 60% faster
- Experimented with text-based features to represent citizen questions and lawyer responses

Software Engineering Intern, Qualcomm
San Diego, CA May 2018 - August 2018

- Achieved over 10% reduction in memory consumption for transceiver chip
- Spearheaded development of tool to eliminate redundancies in radiofrequency settings data for 5G modem
- Modeled embedded software with Abstract Syntax Trees to automate optimization of memory layout

Software Engineering Intern, Synopsys
Mountain View, CA May 2017 - August 2017

- Built Machine Learning API to analyze runtime performance of EDA tools
- Developed web-based push notification system for SQL server

Skills: Python, Tensorflow, PyTorch, NumPy/SciPy/pandas, Transformers, Huggingface, NLTK